

Cultured meat and other alternatives and substitutes to animal products

Masoumeh Niazifar^a, Akbar Taghizadeh^a, Valiollah Palangi^b, Özer Hakan Bayraktar^c, and Maximilian Lackner^d, ^aDepartment of Animal Science, Faculty of Agriculture, University of Tabriz, Tabriz, Iran; ^bDepartment of Animal Sciences, Faculty of Agriculture, Near East University, Nicosia, North Cyprus; ^cDepartment of Animal Science, Faculty of Agriculture, Ege University, Izmir, Turkey; ^dDepartment of Industrial Engineering, University of Applied Sciences Technikum Wien, Vienna, Austria

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Abstract

Globally, red and white meat consumption has increased as a consequence of population growth and changes in lifestyle patterns, with developing standards of living. A major reason for this is the fact that meat is the body's primary source of protein and iron, playing a vital role in human nutrition. Consumer demand for meat is expected to double within the next 20 years, so meat production will have to increase further. As meat production rises, many negative consequences result, such as increased animal slaughter, additional greenhouse gas emissions, excessive water usage, land use (change) and increased disease risks for both humans and animals. As alternative protein ("alt protein"), plant-based produces can be used. Other techniques, such as lab-grown meat production ("cultured meat") and single cell protein (SCP) from bacteria, fungi and algae, have become more and more interesting in the last years, as they eliminate the need to farm and slaughter animals. They offer a route to alternative protein next to said plant protein and insect protein. The first demonstration of cultured meat in the embodiment of a beef burger patty happened in 2013. In the following years, many research labs in academia, as well as in startups, have flourished by diversifying their production to include a wide range of lab-grown meats. Many techniques have been employed to date to cultivate animal meat at a large scale. Even though cell-cultivated meats hold a high promise for scalable and animal-free meat production, they still face economic and technical challenges. As epidemics continue and the worldwide demand for meat grows, meat cultivation can provide us with an essential solution for our long-term nutritional needs. This chapter evaluates whether it is feasible to grow meat in controlled laboratory environments. Using skeletal muscle tissue engineering techniques, stem cells, and concurrent cell and tissue cultivation, meat can be developed in a laboratory. The cultivation of stem cells and their differentiation is regarded as a key approach to producing meat in experimental settings. Cultured meat and SCP are also subsumed as "cellular agriculture." This chapter focuses on cultured meat.

Glossary

Environmental impacts One of the key advantages of cultured meat lies in its reduced environmental footprint. Traditional livestock farming requires huge land areas, water, and fodder to raise animals. Additionally, it contributes significantly to greenhouse gas emissions, deforestation, and water pollution. In contrast, cultured meat production is more resource efficient. It uses less land, requires fewer inputs, and generates fewer greenhouse gas emissions. By minimizing large-scale animal farming, the technology can alleviate some of the environmental pressures associated with meat production.

Ethical and legal issues Despite its promise, cultured meat faces several ethical and legal challenges. Critics argue that the process of extracting cells from animals and growing them in laboratories raises questions about animal welfare and the intrinsic value of life. The long-term effects of consuming lab-grown meat and its impact on human health still have to be fully understood. From a legal perspective, the regulatory framework (e.g., novel food) varies across countries. Some nations have embraced its development, while others remain cautious. Striking a balance between innovation, consumer safety, and ethical considerations is crucial as the industry continues to evolve. Cultured meat products typically require “novel food” approval under various legislations.

Future potential Looking ahead, cultured meat holds significant promise. As the global population grows and demands for protein increase, finding sustainable and scalable food sources becomes essential. Cultured meat could play a vital role in meeting this demand. Researchers are exploring ways to improve production efficiency, reduce costs, and enhance the taste and texture of cultured meat. If successful, it could truly transform the entire food system.

What is cultured meat? Cultured meat, also known as lab-grown meat or cell-based meat, is an emerging alternative to conventionally farmed meat. It is produced by cultivating animal cells in a controlled laboratory environment. In contrast to classic livestock farming, which involves raising and slaughtering animals, lab-grown meat is created without the necessity of raising entire animals. Instead, scientists extract animal cells (usually muscle cells) and place them in a nutrient-rich medium that encourages cell growth. The cells then multiply and form muscle tissue, which is further processed into meat and meat products. The result is a sustainable and potentially more ethical source of protein that aims to address the environmental and ethical challenges associated with conventional meat production. An alternative is single cell protein (SCP), of bacteria, fungi or algae.

Key points

The reader should get an overview of alternatives to classic animal-derived meat. “Alt protein” is not only plant proteins, which are often highly processed and imported, but also can come from cellular agriculture. This chapter gives an up-to-date overview of lab-grown meat and similar techniques, to provide more sustainable meat alternatives and substitutes to a growing world population.

Learning objectives

- Be able to describe the characteristics of lab grown meat.
- Understand the differences to classic farmed animals and alternatives like single cell protein (SCP).
- Be able to describe the pros and cons, and state of development, of cultured meat.
- Be able to understand cellular agriculture concepts.

Introduction

The process of human progress, characterized by an increase in population and demand, has presented difficulties in maintaining sustainable development. Within agriculture, animal breeding has experienced a consistent and positive trajectory. Projections by the FAO indicate that the red meat market is bound to increase 73% by the year 2050, reaching 445 million metric tons/year. Red meat is recognized as a significant dietary source of important proteins and minerals, such as iron, zinc, and vitamin B₁₂ (Jandyal et al., 2021). However, the meat business has faced many obstacles and criticisms in recent years regarding animal welfare and overall condition. Animals for meat a major contributor to greenhouse gas emissions. Conventional meat production is responsible for 15.24% of greenhouse gas emissions (Chriki and Hocquette, 2020). A further rise in red meat production will result in a corresponding increase in land demand, energy consumption, and water usage. Conventional meat production has harmful impacts on the environment, including the depletion of freshwater supplies, soil erosion, loss of biodiversity, and the destruction of habitats (Risner et al., 2020). These variables can greatly exacerbate the inequitable distribution of food on a global scale. The conventional meat production system is a major contributor to greenhouse gas emissions, with carbon dioxide accounting for 9%, methane for 39%, and nitrogen oxide for 65% of the sector's emissions (Bryant and Barnett, 2018). Additionally, this system requires considerable land use and consumes excessive amounts of water and other resources. Rectifying conventional meat production

results in substantial amelioration of environmental pollutants. Research indicates that the meat production system in controlled laboratory environments can decrease energy use by up to 40%, land use by 99% and water consumption by 90% compared to traditional meat production (Bhat et al., 2019).

White meat (e.g., chicken) and fish (aquaculture) have a better feed conversion ratio than cattle, still they are not fully sustainable and scalable. The idea of synthetic meat dates back into the 19th century. Among the early proponents of the idea of cultured meat is Winston Churchill, a British author, and politician, who mentioned cultured meat production in a 1931 article (Li et al., 2015). In 2013, a Dutch company, MosaMeat with Prof. Mark Post, successfully constructed lab-grown beef and subsequently manufactured the inaugural lab-made hamburger. The production cost of the hamburger amounted to approximately \$325,000/kg of meat. Subsequently, scientists have endeavored to reduce the expense associated with the creation of lab-grown meat. One can create laboratory-grown meat, often called cultured meat, by cultivating live cells in a laboratory setting (Chriki and Hocquette, 2020). Living cells are collected and grown in a controlled setting to yield lab-grown meat, resulting in the creation of meat without the need to slaughter animals. This method involves extracting a muscle-producing stem cell from the animal's body and then stimulating and multiplying it through laboratory techniques. The production of lab-grown meat is fundamentally a straightforward process, as it closely resembles the techniques used to cultivate and multiply various cells, a procedure that scientists have employed since 1900. Nevertheless, it is impractical to compel the entire global population to cease consuming meat (Collins et al., 2007). Cell cultivation has evolved as a new technology in response to the growing need for meat production and the need to solve issues such as environmental degradation and animal welfare. It successfully rolled out to replace significant amounts of classic meat, this can constitute a "food technology revolution," as it completely changes the way of meat production. Laboratory-grown or cell-cultivated beef—and other meat—is a promising avenue to explore. Nevertheless, it is imperative that the alternative meat closely resembles conventional meat and guarantees human consumption safety. Lab-grown meat is also called clean meat (McDonald and Vivalt, 2018), a term potentially more appealing to consumers. It alludes to lower levels of emissions per kg of meat. Surveys indicate that the majority of the general public has limited enthusiasm for consuming synthetic meat. A recent article in the journal "PlosOne" revealed that over 60% of respondents expressed a willingness to sample such a product at least once. However, only one-third of individuals expressed no reservations about consuming lab-grown meat (Bonnet et al., 2020).

To lab-cultivate meat, scientists extract a cow or other cells from the tissue without causing harm to the animal. Subsequently, under carefully controlled laboratory conditions, these cells undergo growth and multiplication to generate muscle tissue. Most of the meat people consume daily consists of muscle tissue (Bryant et al., 2019).

What is needed to grow cultured meat:

1. Primary cells: To cultivate meat in the laboratory, it is essential to first use a population of cells that show rapid growth and division. Stem cells and myoblasts cells are highly sought after. Stem cells show rapid rates of division, yet their ability to effectively differentiate into muscle tissue remains a significant barrier in terms of regulation and promotion. Mature muscle cells, which possess the ideal structure of muscular tissue, by contrast, have significantly lower growth and division rates.
2. Growth medium: Cells require a variety of nutrients to proliferate and form the muscle tissue of the meat (Post and Hocquette, 2017). The medium is a cost driver.

The growth medium for cultured meat production is a critical factor influencing cost and quality. It requires various nutrients, growth factors, and hormones to support cell proliferation and differentiation (Ahmad et al., 2023; Lee et al., 2024). Key components include insulin-like growth factors, fibroblast growth factors, and hormones such as insulin and testosterone (Ahmad et al., 2023). The medium must be food-grade, cost-effective, and animal ingredient-free while regulating large-scale cell growth and differentiation (O'Neill et al., 2021). Challenges in cultivated meat production include consumer acceptance, scalability, safety, and reproducibility (Santos et al., 2023). The process involves selecting appropriate cell sources, providing necessary nutrients and signaling molecules, and using scaffolds or microcarriers for three-dimensional organization (Lee et al., 2024; Santos et al., 2023). Optimizing media formulations by applying principles from large-scale microbial fermentation and conventional cell culture can help address these challenges (O'Neill et al., 2021).

3. Cellular scaffolding: To develop into three-dimensional tissue, cells require a framework on which to grow. Cells need a biodegradable frame capable of absorbing them once the flesh tissue is produced. From a biological standpoint, lab-grown beef does not differ from conventional meat. During culturing, scientists propagate and multiply cells within filamentous or circular frameworks (Burgat et al., 2022). Scaffolding plays a crucial role in cultured meat production, significantly influencing the final texture and haptic properties of the product (Nurul Alam et al., 2024). Various scaffolding technologies, including fiber scaffolds, hydrogels, and micro-carrier beads, are essential for mass production and can be tailored to mimic the appearance, texture, and flavor of conventional meat (Lee and Choi, 2024). The choice of scaffold materials and manufacturing methods directly impacts the texture characteristics of the final product (Bomkamp et al., 2022; Lee and Choi, 2024). Researchers are exploring edible, low-cost, and sustainable materials for scaffolding to ensure food safety and scalability (Levi et al., 2022). Advanced techniques like 3D scaffolds, 3D printing, and electrospun nanofibers are being implemented to replicate the fibrous microstructure of muscle tissues and enhance sensory attributes (Nurul Alam et al., 2024). While challenges remain, ongoing research in scaffolding technologies offers opportunities to customize meat properties and improve the overall quality of cultured meat (Bomkamp et al., 2022).

A brief timeline of lab-grown meat's development

Let us analyze the historical context of cultured meat and then shed light on the latest technological progress and current challenges. Cultured meat has long been envisioned, but a method to achieve it had not been developed. In 1897, the concept of cultured flesh was brought up in a science fiction book titled “Auf Zwei Planeten” (“On two planets,” Kurd Laßwitz) and later appeared in various other books during the past century, such as “Ashes” (Barjavel, 1943). In 1907, researcher Ross Harrison successfully cultured frog nerve cells in lymph medium at Johns Hopkins University (Lindsay et al., 2024). In 1931, it was Winston Churchill who made a significant forecast, possibly influenced by his associate Frederick Edwin Smith. He expressed the intention to refrain from the irrational habit of raising a whole chicken just to consume its breast or wings. Instead, Bhat et al. (2017) proposed cultivating these specific parts separately using a proper medium. Using synthetic food is certain to occur in the future (Bhat et al., 2017). For a considerable period, laboratory-based scientific research has included in vitro approaches, such as tissue cultivation, to address wounds and diseases. However, cultured meat production made little progress until the late 20th century. Willem van Eelen submitted the initial patent for a technique to produce cultured beef in the late 1990s (patent number US7270829). In 1998, NASA cultivated goldfish meat in a laboratory as part of research aimed at extended space voyages (Hildreth, 2023). The initial consumption of produced flesh by humans seems to have been facilitated by the artist Oron Catts. During his 2003 experiment “Disembodied Cuisine,” he successfully cultivated cells from a frog and showcased them as miniature frog steaks at a museum supper in the French city of Nantes (<https://tcaproject.net/portfolio/disembodied-cuisine/>).

Two decades ago, the government of the Netherlands allocated funding for research programs dedicated to cultured meat development. Additionally, Mark Post and Google cofounder Sergei Brin accelerated its development (Collins et al., 2007). Researchers recorded the first scientific publication on cultured meat in 2008 (Guan et al., 2021). In that year, the Norwegian Food Research Institute held a symposium specifically focused on the topic of cultured beef. Public awareness greatly went up during a televised event in London in 2013, where Prof. Post demonstrated the preparation and serving of cultured beef. Two journalists publicly tasted the meat and declared that it closely resembled conventional meat (Jandyal et al., 2021; Wilks and Phillips, 2017). In December 2020, a restaurant in Singapore celebrated the maiden sale of a cultured beef product after it had obtained novel food approval (Ye et al., 2022).

New Harvest (<https://new-harvest.org/>) and the Good Food Institute (GFI, <http://www.gfi.org>), two non-profit organizations, have helped to provide support of alt protein in general and cultured meat in particular. Startup enterprises are conducting a substantial amount of cutting-edge research in the field. Presently, there are more than 50 startups dedicated to cultured meat and related fields such as artificial milk or seafood. Israel, the USA and the EU host the bulk of startups, while Asia has smaller presences, particularly in Singapore. Private individuals, including notable billionaires like Li Ka-shing, Bill Gates, and Richard Branson, constitute the bulk of investors (Guan et al., 2021). Furthermore, the traditional meat market is home to well-established firms, such as Cargill and Tyson Foods in the US and PHW (Lohmann & Co. AG), Migros, and Grimaud in the EU (Lindsay et al., 2024).

In 2013, researchers created the first in vitro meat burger, marking a pivotal turning point in the development of cultured meat technology. The burger consisted of a five-ounce lab-grown beef patty, which was cooked and evaluated by a panel of sensory experts in London. The judges concluded that its flavor closely resembled that of a conventional beef burger. The meat was cultivated over the course of 3 months using stem cells extracted from a cow's shoulder. The total cost of producing the burger exceeded \$330,000/kg of meat. This landmark event generated widespread public interest—particularly among consumers concerned with animal welfare—and spurred demand for the commercialisation of cultured meat products (Zaraska, 2013). In recent years, investment in alternative protein companies has surged, with the global alternative protein sector projected to reach \$290 billion (Cullen, 2021). Nevertheless, investment in cultured meat startups remains relatively limited compared to plant-based enterprises, despite notable increases in funding (Dolgin, 2019). Investment in the agrifood sector has also grown, particularly through equity-related funds, resulting in shared ownership of major agrifood corporations by asset management firms (Clapp, 2019). This concentration of market power may have significant implications for the broader food system.

Moreover, disparities in research funding persist, with major commodity crops receiving three to five times more support than underutilized crops such as grain legumes and coarse cereals (Bollington et al., 2021). These investment patterns pose challenges to food system transformation and underscore the need for increased public investment in diverse crops, along with cross-sector collaboration to promote nutritious diets and planetary health (Bollington et al., 2021). While the investments in question have indeed seen a notable increase in recent years, they nevertheless remain comparatively low when compared to investments in firms focused on plant-based food (Bryant et al., 2019), such as Beyond Meat and Impossible Foods. Plant-protein is low cost, yet the products are often criticized for being highly processed.

The approach used to create cultured meat is based on tissue engineering. Previously, researchers have primarily focused on the medical applications of this technique, specifically in the fields of regenerative medicine, toxicology, and drug discovery, where non-animal in vitro models are used (Jandyal et al., 2021). Entrepreneurs in this field face challenges in determining the specific areas of concentration due to the yet-to-be-achieved industrial-scale manufacturing of cultured meat. The primary technological hurdles include the development of suitable cell lines, serum-free growth media, scaffolding materials, and efficient bioreactors (Gu et al., 2025; Zheng et al., 2025). Producing stable meat products with desirable texture, flavor, and aroma remains a persistent obstacle (Chodkowska et al., 2022). In addition, regulatory barriers and the need for consumer acceptance further complicate commercialisation efforts (Stephens et al., 2018). Despite these challenges, cultured meat has been approved in Singapore and the United States, marking a step forward toward market viability (Chodkowska et al., 2022). Researchers are exploring innovative

strategies to reduce production costs, improve product quality, and address environmental concerns (Zheng et al., 2025). As the field advances, interdisciplinary collaboration will be crucial in overcoming technical, socio-political, and regulatory challenges to enable large-scale cultured meat production (Gu et al., 2025; Stephens et al., 2018). Researchers have recognized the fundamental principles. Stem cells are derived from muscle tissue or embryos and undergo a process of expansion and differentiation to ultimately become muscle cells. Subsequently, these cells are introduced into a bioreactor to undergo proliferation and are then moved to a matrix or scaffold to help with the development of muscle fibers and larger tissue. The growing medium is an important factor in cost and uncertainty (Singh et al., 2022), hindering so far the reaching of price parity with meat.

Lab-grown meat research and development currently encounters some technical obstacles. The production of minced meat, burgers and nuggets should probably occur well in advance of steak. Undoubtedly, the process of cultivating fully developed muscle tissue, complete with its intricate network of membranes and filaments, requires a sophisticated system involving several types of cells that grow in a coordinated way, as well as a structure that replicates a network of blood vessels. Acquiring meat marbling¹ can be quite challenging. Furthermore, the methods for replicating meat's authentic color and tenderization² through in vitro processes are still uncertain, as they do not currently mimic the natural maturing process. Regarding a minced product, the scale differs from that of a full-thickness meat product, particularly at the last stage of organoid or tissue synthesis.

Therefore, the methods used to generate cultured meat are not yet advanced enough to accurately replicate the wide range of flavors and textures seen in different meat products. Furthermore, future producers will use uncertain cell types and growing media. However, it is quite likely that the procedure will not depend on animal-derived substances, such as the fetal bovine serum³ utilized in the initial phases of the growth media or the cow collagen⁴ often employed in the scaffold today. However, substantial advancements in technology are still required to make cultured meat production economically viable and feasible for large-scale food production. These advancements encompass various aspects, including cell selection, medium optimization, biomaterials, and tissue engineering (Lewisch and Riefler, 2023).

Meat cultivated in in-vitro conditions

Laboratory-grown meat is a promising alternative to conventional meat such as beef or pork, addressing environmental and ethical concerns associated with traditional livestock farming. Particularly the beef industry significantly contributes to greenhouse gas emissions and resource depletion (Grass, 2019). Lab-grown meat, produced through cellular agriculture, offers potential solutions to these issues (Sergelidis, 2019). It requires fewer resources, reduces the risk of pathogen contamination, and allows for nutritional content regulation (Roy et al., 2021). This technology could help meet the growing global demand for meat while mitigating environmental impacts (Behera and Adhikary, 2023). However, challenges remain, including higher energy consumption, potential long-term health effects, and cultural acceptance (Roy et al., 2021). Despite these obstacles, lab-grown meat presents a sustainable alternative to conventional beef production, potentially addressing concerns related to animal welfare, environmental sustainability, and food security (Behera and Adhikary, 2023; Grass, 2019). Recent advancements have led to the development of several main procedures, including the separation and identification of stem cells. These procedures include cell culture and tissue engineering. These approaches can generate skeletal muscles and connective tissues. Medical implants use biological artificial muscles generated from static stem cells, skeletal muscles, and satellite cells. Moreover, artificial muscles are considered a rich source of animal protein. A bioreactor is required for stem cell cultivation and differentiation. Tissue culture can produce edible animal muscle, commonly referred to as meat (Singh et al., 2022). This process involves the significant expansion of muscle cells to form a substantial tissue or mass of muscle cells. The development of laboratory-grown meat offers a promising avenue for obtaining animal-free meat, representing a novel approach. Meat studies and testing are slow because of their high expenses. The cost of producing lab-grown meat, in the very early days, was as high as \$50,000 to even \$500,000 per kg, and it required cultivation in small quantities (Vialles, 1988). Current estimates range from \$17 to \$63 per kg for large-scale production (Garrison et al., 2022; Negulescu et al., 2023), which is still far from price parity with conventional meat. The main cost drivers are cell culture media, bioreactors, and labor, accounting for over 80% of production expenses (Garrison et al., 2022). To address these challenges, researchers are exploring systems biology and metabolic modeling approaches to optimize cell culture media and reduce costs (Gomez Romero and Boyle, 2023). Scaling up production using larger bioreactors, such as 262,000 L airlift reactors, could potentially lower costs to \$17/kg (Negulescu et al., 2023). However, achieving price parity with traditional meat products remains a significant hurdle, and the industry may need to approach technical limits for profitability as a commodity (Risner et al., 2020). Despite these challenges, scientists remain optimistic about the prospects of cultivating meat in controlled laboratory environments. Despite these challenges, scientists remain optimistic about the prospects of cultivating meat in controlled laboratory environments.

¹The intramuscular fat content in meat.

²Meat tenderization is a crucial process in the meat industry, aimed at improving sensory attributes like tenderness and palatability.

³A widely used supplement in cell culture media, providing essential growth factors for various cell types.

⁴Cow collagen has been extensively studied for its biomedical applications and potential risks. Research has identified dermal dysplasia in cows characterized by collagen disorder-related skin fragility. Human immune responses to bovine collagen have been observed, with antibodies reacting to various collagen fragments.

Possible methods of meat production in laboratory conditions

Skeletal muscle cells

Under laboratory conditions, the presence of muscle cells and proteins is essential for the production of meat. Potential sources of meat synthesis under laboratory conditions include structural and loose muscle cells. When precursor cells fuse, they replicate the multinucleated muscular fibers that compose the muscle. Embryonic stem cells, satellite cells, and specialized cells inside muscle tissue all have the potential to serve as a valuable reservoir for generating muscle cells. In recent decades, researchers have employed myoblasts and satellite cells for cultivating meat. After being cultured, the satellite cells show significant differences between myotubes and myofibers (Niazifar et al., 2024). Additionally, a limited number of satellite subsets with superior reconstruction capabilities are favored for skeletal muscle tissue engineering purposes (Jandyal et al., 2021). Researchers are exploring plant, microbial, and fungi-derived compounds as alternatives to animal-based media components (Kumar et al., 2024). Co-culturing various cell types, such as adipocytes and fibroblasts, is being investigated to improve texture and organoleptic properties (Kumar et al., 2024).

Stem cells

Stem cells possess distinct characteristics that separate them from other cells. Stem cells can reside inside a reasonably defined location, and there have been multiple stem cells identified to cultivate meat in laboratory settings. Before implantation, researchers obtain fetal stem cells from the fetal reproductive system (Singh et al., 2022).

Induction power stem cells

One can regard pig-induced pluripotent stem cells as an alternative means of cultivating meat. Bio-artificial muscles are created using pig-induced stem cells, which differentiate into myotubes and aid in intraocular muscle injury restoration. The cellular cytoskeleton does not possess a mechanism for generating meat in laboratory settings. Consequently, the cells are undeveloped, and the protein origin is not commonly known as the primary myotube. Early myotubes differentiate into adult cells following myoblast fusion. These early myotubes then undergo secondary differentiation to provide enough biomass for meat production under laboratory conditions (Singh et al., 2022).

Scaffolding techniques

While small muscle fragments can get adequate nutrients and oxygen through diffusion, there has been little progress in creating cultured muscles that have integrated blood vessels for oxygen and nutrient supply (Bhat et al., 2017). The primary obstacle to in vitro meat production lies in determining the ideal composition of the growth medium. This is despite the availability of several cell culture techniques. The ideal medium should be cost-effective, composed primarily of food-grade materials, easily obtainable in large amounts, and efficient in promoting and supporting the growth, reproduction, and specialization of muscle cells.

Biophotonics

Biophotonics is an emerging technique that uses laser light to effectively join particles. By creating specific structures, biophotonics generates “optical matter” where material can be placed and maintained until the light dissipates. The constituent elements can amalgamate to create a novel, cohesive framework (Palshikar et al., 2013). It encompasses various applications, from molecular-level interactions to whole organism imaging (Tsia, 2016). Femtosecond lasers have enabled significant advancements in cell manipulation, including transfection and cell fusion, by exploiting their ability to open cell membranes with minimal phototoxicity (Tirlapur and König, 2002). Recent developments have introduced the concept of optobiology, where living cells can serve as optical components such as microlenses, microresonators, or waveguides (Miccio et al., 2020). We still lack a comprehensive understanding of the underlying mechanisms responsible for this remarkable characteristic of light. The revolutionary technology has the potential to produce meat by fusing muscle cells. It has the advantage of easily incorporating traits like fat, in contrast to conventional methods. This innovative approach allows for diverse applications in imaging, soft matter manipulation, and biomedical diagnostics. Biophotonics continues to revolutionize life science studies and medical applications, offering new possibilities for understanding and treating diseases through the interaction of light with biological tissues (Palshikar et al., 2013; Tsia, 2016). Biophotonics may serve as a substitute for standard scaffolding techniques in cell manipulation. Researchers have successfully employed biophotonics to generate red blood cells and hamster ovaries (Javaid and Haleem, 2019).

Self-organization techniques

Another approach to obtain highly structured alternative meat includes using surgically removed animal muscle tissue. It entails the development of self-organized muscle tissue or the expansion of pre-existing muscle tissue in a controlled laboratory environment. Benjaminson et al. (2002) investigated the capacity of developed muscle tissue cells to attach to and multiply on a surface.

The researchers divided goldfish tissue samples into sections and then exposed them to centrifugation to create pellets. Subsequently, the researchers introduced these pellets into Petri plates filled with a nutritional solution and subjected them to a 7-day incubation period. Benjaminson et al. (2002) conducted a study to examine the impact of several growth mediums, including fetal bovine serum, fish meal extract, and various mushroom extracts, on the growth of explant muscle tissue. The aim was to determine the distinct advantages of each growth medium and maybe identify substitutes for fetal bovine serum. After 14 days of cultivation, 81% of the 48 cultures demonstrated tissue adhesion to the culture vessel, 63% showed self-healing skills, and 74% displayed cell proliferation. The excised tissue showed a growth rate of over 14% when nourished with fetal bovine serum and a growth rate surpassing 13% when nourished with maitake mushroom extract (Pilgrim et al., 2022). After 1 week of culture, the goldfish skeletal muscle cells experienced a substantial 79% growth, as evidenced by a significant increase in surface area. Before undergoing deep-frying, the explants and newly generated tissue were submerged in olive oil and garlic, and then submitted to a sensory panel for evaluation (Bhat et al., 2019). The sensory panel noted that the explants and freshly formed tissue had visual and olfactory attributes that indicated their suitability for ingestion. Guan et al. (2021) devised a method for extracting and swiftly duplicating pig muscle cells. They divided the muscles into smaller fragments and then subjected them to centrifugation to separate the cells and generate pellets. The pellets were then cultivated on Petri dishes using a growth mixture containing fetal bovine serum and penicillin-streptomycin to enhance cell proliferation. Furthermore, horse serum was added to the differentiation medium to stimulate cellular differentiation. The proliferation assessment reveals a proliferation rate of over 70% during 1 week. Wang et al. recently acquired goat skeletal muscle cells and achieved an 80% increase in their quantity by utilizing a growth medium comprising fetal bovine serum together with a differentiation medium that included horse serum. The aforementioned research showcases the effectiveness of fetal bovine serum and horse serum in stimulating the growth and proliferation of muscle cells. Research is essential for developing precise techniques that use chemicals to target various types of animal muscle cells to produce lab-grown meat using the self-organizing method (Post et al., 2020).

Culture media and growth factors

An ideal culture medium should support and promote growth while being easily obtainable, cost-effective, and consumable. Cell growth relies on the presence of important nutrients such as amino acids, fatty acids, vitamins, trace minerals, and extracellular vesicles in the media. In addition to antibiotic/antimitotic combinations, specific cultures require embryo extract. Muscle cells serve as the main producers of insulin-like growth factors and are essential for the production of lab-grown meat. Myoblast differentiation and fusion can be enhanced by reducing the quantities of mitogenic growth factors, a technique often employed by scientists. After cell proliferation, an insulin-like growth factor is generated, leading to differentiation and the creation of myotubes. The specific serum components that are principally responsible for cell growth remain uncertain, despite the involvement of several growth factors, inhibitors, and metabolic moderators. The scaffold composition, like the culture media, is associated with the generation of in vitro meat. A wide range of biomaterials have undergone testing. Animal-derived scaffolds, like collagen, resemble the natural microenvironment in living organisms. Specialized muscle cells, known as myoblasts, opt to arrange themselves in a parallel way, apply pressure, and ultimately develop into muscle fibers. The most effective endeavors to produce lab-grown meat have utilized scaffolding made of collagen, whereas tries to use artificial biomaterials have faced challenges in tissue contraction (Welin, 2013).

3D/4D organ or bio-printing

Additive manufacturing as 3D or 4D organ “bioprinting” relies on standard printing principles. 3D bioprinting is an emerging technology with significant potential in regenerative medicine and organ fabrication (Cui et al., 2017; Vijayavenkataraman et al., 2018). This technique enables the fabrication of complex tissues composed of multiple cell types, offering precise control over their composition, spatial distribution, and structural organization. While the complete bioprinting of functional organs remains a long-term goal, notable progress has been made in printing various tissue types, including skin, bone, cartilage, and cardiovascular tissues (Ali et al., 2018; Loai et al., 2019). CAD software is used to generate a bioproduct prototype. Using computer-aided design (CAD), researchers apply cells to gels, and through cultivation, the cells merge to create the bio-product. The bio-product possesses the fundamental biological composition and develops blood vessels for blood delivery. 3D bioprinting is an efficient and attractive method for producing organs or tissues that are both functionally and anatomically equivalent, specifically for regenerative tissue and organ therapy. It precisely places biomaterials and other types of cells into a unified 3D tissue structure. 4D printing, utilizing similar technology, expands upon the concept of 3D printing by introducing an 4th dimension, which is that of temporal change (Javaid and Haleem, 2019) next to x-, y- and z-spatial dimensions. This method restores muscle, bone, and cardiovascular tissues using the sensitivity of the target organs or tissues to both humidity and temperature (Jandyal et al., 2021). Aleph Farms achieved the milestone of producing the world’s inaugural rib-eye steak through the use of 3D bioprinting in 2021. The steak has fat comparable to conventional beef and is tender and succulent. The company also asserts its capability to make various types of meat using this technology in the future (Srutee et al., 2022). The technique offers several advantages, including patient-specific design capabilities, high resolution, and adaptability (Cui et al., 2017). However, several challenges remain, such as improving scalability, developing advanced bioinks, and addressing ethical and regulatory issues (Vijayavenkataraman et al., 2018). Despite these obstacles, 3D bioprinting holds considerable promise for the fabrication of functional tissues and organs, potentially revolutionizing transplantation practices and alleviating the global shortage of donor organs (Ali et al., 2018; Vijayavenkataraman et al., 2018).

Bioreactor

Previous studies have examined the significance of bioreactor configurations for tissue regeneration. Researchers widely employ static bioreactors, where cells are placed on a scaffold, followed by the addition of appropriate growth media. They are then incubated in a controlled environment. The manufacture of in vitro meat will require the creation of innovative bioreactors that stimulate tissue growth while retaining huge volumes of low-shear and homogeneous perfusion.⁵ Dvir et al. (2006) developed a novel perfusion bioreactor with a flow-distributing mesh and open-pore fixing nets, enabling homogeneous fluid flow and maximal construct exposure to medium. Nazempour and Van Wie (2018) introduced a bioreactor combining shear stress and oscillating hydrostatic pressure to enhance cartilage tissue growth. Guyot et al. (2016) created a 3D computational model using the level set method to simulate neotissue growth in a perfusion bioreactor, considering scaffold geometry and flow-induced shear stresses. Osborne et al. (2010) examined a three-phase model for tissue construct growth in a perfusion bioreactor, comparing cell density-, pressure-, and shear stress-regulated growth. Skeletal muscle tissue engineering research has made extensive use of rotating bioreactors (Ge et al., 2023).

Bioprocesses are comprised of four distinct stages: proliferation (rapid cell multiplication), cellular differentiation (development into specialized cell types), product synthesis (generation of desired biomolecules or tissue), and waste utilization (management or conversion of metabolic byproducts). The biomechanical forces contribute to the formation of a smooth and uninterrupted flow of the substance. Cell proliferation and differentiation are fundamental processes in stem and progenitor cells, with metabolic pathways playing a crucial role in supporting these functions (Agathocleous and Harris, 2013). Surface topography significantly influences cell behavior, including adhesion, migration, and differentiation, which is essential for various biomedical applications (Tudureanu et al., 2022). In microbial processes, cell metabolism and product formation are key aspects of process development, along with downstream processing and waste utilization (Doelle, 1994). Plant cell fermentation for secondary metabolite production involves cell differentiation, product synthesis, and accumulation, with industrial-scale operations being well-established (Constabel, 1988). However, the induction of product formation and analysis of cell competence for product formation when associated with structural differentiation remain areas of ongoing research, potentially requiring genetic engineering techniques (Constabel, 1988). Understanding these processes is crucial for advancing tissue engineering applications and developing sustainable microbial processes. This enhances the process of diffusion and achieves a high rate of mass transfer while minimizing the level of shear stress. Direct-perfusion bioreactors are more suitable for scaffold-based cultivation. In this situation, the medium flows via a permeable scaffold, while gas exchange takes place in an external fluid loop. This particular type of bioreactor has a high rate of mass transfer and a notably low level of shear stress (Vergeer, 2021).

Why do we need meat alternatives?

There are at least three reasons to encourage the investigation of alternate methods to cattle and other traditional production processes for classic meat. Initially, due to the anticipated substantial rise in demand for meat, we will rapidly deplete our ability to produce enough. A major portion of cultivable land is already allocated to cultivating animal feed and maintaining cattle, which primarily causes this issue. Research indicates that approximately 12% of the Earth's ice-free land surface is used for cropland, while 22–27% is utilized for pasture and rangelands (Leff et al., 2004; Ramankutty et al., 2008). Agriculture currently occupies approximately 30% of the global land area, significantly expanding into forests and ecologically diverse regions (D'Silva, 2017). Corn and soy, primarily grown for livestock feed, are major drivers of deforestation and soil degradation (D'Silva, 2017). Modern intensive dairy farming relies heavily on grain and pulse concentrates, which are potentially edible for humans, leading to competition for arable land (Knaus, 2013). Furthermore, there is significant apprehension regarding livestock production and management's ecological impact. The sector contributes to greenhouse gas emissions, land degradation, water pollution, and biodiversity loss (Bezabih and Berhane, 2014; Ussiri and Lal, 2013). Furthermore, the practice of herding and the large-scale slaughter of animals has given rise to significant societal concerns regarding both animal welfare and public health (Zeng, 2023).

The benefits of laboratory cultivation meat

When cultivating laboratory meat, using growth hormones is unnecessary, resulting in a final product that is distinct from hormone-infused meat and therefore healthier. Moreover, allocating a staggering 80% of the global supply of antibiotics for human and animal health purposes leads to a surge in antibiotic resistance among humans. Consuming feed containing pesticides and chemicals can contaminate animal meat (Chriki and Hocquette, 2020). However, laboratory-cultivated meat significantly reduces contamination risk. Cultivated laboratory meats have the potential to be entirely devoid of fat. Furthermore, if it is intended to maintain the texture and flavor. To ensure consumers' health, these meats contain a certain amount of fat that can be produced without cholesterol and enriched with omega-3. Regulating the quality assurance of these meats to ensure they are devoid of

⁵Perfusion is the continuous flow of nutrient-rich growth media through a tissue or scaffold to ensure adequate nutrient delivery, waste removal, and cell viability in bioreactors.

parasites and common diseases that can be transmitted between humans and animals is more efficient, convenient, and cost-effective (Javaid and Haleem, 2019).

Sustainability

We define sustainability as the capacity to maintain a particular level of activity or progress over an extended period of time without inflicting damage on the environment or exhausting resources. Animal production, first and foremost beef, occupies around 70% of the entire agricultural land (Lindsay et al., 2024). However, there is a continuous discussion regarding the exact environmental repercussions. According to certain writers, cattle are responsible for about 18% of the total greenhouse gas emissions (Collins et al., 2007). Ruminants, in particular, are the primary sources of methane emissions, accounting for about 37% of the entire emissions (FAO, 2006). Nevertheless, certain authors have presented far lower figures (FAO, 2006), and a consensus among specialists has yet to be reached. Methane emissions from ruminant livestock significantly contribute to global anthropogenic greenhouse gas emissions, accounting for 5–6% worldwide (Clark, 2013). Ruminants are significant contributors to global methane emissions, with estimates ranging from 17% to 37% of anthropogenic sources (Sejian and Naqvi, 2012; FAO, 2006). Methane production in ruminants is influenced by various factors, including body mass, diet, and animal species (Pérez-Barbería, 2017; Broucek, 2014). Cattle generally produce more methane than other ruminants, with dairy cows emitting 151–497 g/day and beef cattle 161–396 g/day (Broucek, 2014). To enhance the effectiveness of environmental sustainability objectives, it is recommended to reduce the overall population of ruminant animals, as they are the primary contributors to greenhouse gas emissions in the meat industry (). Reducing methane emissions from ruminants remains technically challenging, especially for grazing animals, and must be balanced against increasing demand for animal products and maintaining farm profitability (Clark, 2013). However, global estimates of methane emissions from wild ruminants are lower (1.094–2.687 Tg/year) than previously reported (Pérez-Barbería, 2017). The rumen microbiome plays a crucial role in methane production, with specific microbial communities associated with high or low methane emissions (Tapio et al., 2017). Accurate global estimates are hindered by uncertainties in wild ruminant population sizes (Pérez-Barbería, 2017). The rumen microbiome plays a crucial role in methane production, with specific microbial communities associated with high or low methane emissions (Tapio et al., 2017). Cultured meat products can be more climate-friendly than traditional meat production. However, it is important to do a comprehensive lifecycle analysis to support this claim. Takahashi and Yamanaka (2006) conducted a surface-level life cycle analysis and discovered that under specific production conditions, lab-grown meat has the potential to significantly decrease energy consumption and land usage by 99%, water usage by 90%, and energy consumption by 40% compared to traditional meat production. If implemented at large scale, these reductions could result in a significant decrease in greenhouse gas emissions (Philippon, 2018). The bulk of meat substitute products currently available on the market are made using alternative protein sources, such as soy, wheat proteins, or mycoprotein. The amount of plant biomass required to create 1 kg of meat varies depending on the animal species and other factors, ranging from 2 to 15 kg. For corn, soybeans, alfalfa, and switchgrass, the cumulative energy requirement ranges from 0.97 to 2.66 MJ/kg, with associated global warming impacts of 89–286 g CO₂ equivalent/kg (Kim and Dale, 2005). Theoretically, 1.34 g of glucose is needed to synthesize 1 g of maize plant biomass in darkness, consuming 0.14 g of oxygen and producing 0.24 g carbon dioxide (De Vries et al., 1983). In ruminant production, energy input per kilogram of milk ranges from 1 to 10 MJ, with varying nutrient losses (Tamminga and Verstegen, 1992). Cattle are particularly resource-intensive, requiring 8 kg of feed to produce 1 kg of meat (Huis et al., 2013). For beef production, one study found that 52 kg of food from six plant species was needed to produce 1 kg of boneless meat over 50 months (Fortes et al., 2020). In contrast, insects are highly efficient, converting 2 kg of feed into 1 kg of insect mass (Huis et al., 2013). Broiler chickens, laying hens, and dairy cows are more efficient in producing edible protein per kilogram of body weight compared to beef cattle (Flachowsky, 2017). Global crop allocation for animal feed is a significant concern, with studies indicating that approximately 36–40% of worldwide crop calories are used as livestock feed (Cassidy et al., 2013; Pradhan et al., 2013). This allocation has substantial implications for food security and environmental sustainability. Pradhan et al. (2013) found that about 4 kcal of crop products are required to produce 1 kcal of animal products globally, though this ratio varies widely across regions. The demand for animal feed is projected to nearly double by 2050, driven by population growth and changing dietary patterns (Pradhan et al., 2013). Cassidy et al. (2013) suggest that redirecting crops from animal feed to direct human consumption could potentially increase available food calories by up to 70%. However, Le Cotty and Dorin (2012) predict a more moderate increase in crop needs for livestock, ranging from 8% to 117% of current levels by 2050, depending on dietary scenarios and regional differences in production efficiency. Various studies provide evidence that the intake of plant proteins results in a direct decrease in total carbon emissions when compared to the consumption of meat. Nevertheless, livestock, especially ruminants, can ingest agricultural waste and can flourish on terrain unsuitable for crop cultivation. Therefore, they are highly suitable for the adoption of agroecological systems.

Environmental benefits

A study conducted at the University of Amsterdam reveals that lab-grown meat has the potential to yield the same amount of meat while using only 2% of the resources required by the traditional cattle business (Sebo, 2018). In addition, lab-grown meat will reduce greenhouse gas emissions throughout the process by 50% compared to traditional industries, as well as energy usage. Furthermore, altering the utilization of land per hectare for cattle grazing poses a threat to the survival of 10 indigenous creatures in the region (Barry and Huntsinger, 2021). By requiring significantly fewer resources for the production of lab-grown meat, we can

ensure the preservation of millions of indigenous organisms. Substituting synthetic meat for organic beef has the potential to decrease greenhouse gas emissions by as much as 5% (Lusk et al., 2022). The process of dismantling industrial livestock operations has a substantial impact on minimizing the deforestation and degradation of global forests and rangelands. Under such conditions, there is no necessity to eradicate trees and cultivate fodder to nourish the livestock. Additionally, this technology could be beneficial for the production of lab-grown meat for animals. The unsatisfactory circumstances in which the animals are kept and their susceptibility to diseases such as bovine spongiform encephalopathy and smallpox result in significant expenses for the livestock business (Takahashi and Yamanaka, 2006). Additionally, the notable advantage of lab-grown meat is its suitability for incorporation into the human body during the manufacturing process. Statistical data related to cattle attribute a significant proportion of the phenomenon known as global warming (Gerber et al., 2013). Currently, about 80% of cattle worldwide experience heat stress for at least 30 days annually, with projections indicating increased severity and duration under various climate scenarios (North et al., 2023). Global cattle populations have grown, especially in developing countries, correlating strongly with human population growth and methane emissions (Kozicka et al., 2023). Livestock and cattle generate approximately 18% of greenhouse gas emissions. The emission of greenhouse gases, particularly methane gas, from cow manure has a substantial influence on global warming. However, the crucial aspect of this approach is that this product does not necessitate genetic tinkering, hence mitigating the potential hazards associated with meat consumption.

Health and safety aspects

Proponents of in vitro meat argue that producing it in a controlled environment without other organisms enhances its safety compared to traditional meat, which is exposed to the external world, despite the protection provided by the skin and/or mucosa of each tissue. Cultured muscle cells are not exposed to potential contamination during slaughter because they lack digestive organs nearby, unlike conventional meat. Consequently, they lack the opportunity to contact intestinal pathogens such as *E. coli*, *Salmonella*, or *Campylobacter*. These three pathogens are responsible for the majority of foodborne illness cases annually (Guan et al., 2021). However, scientists or manufacturers may argue that they are never fully capable of complete control. Any error or omission could have severe repercussions for a health issue. This phenomenon is now common in the current era of industrialized meat processing (Javaid and Haleem, 2019).

By not being derived from animals reared in a restricted environment, cultured meat eliminates the risk of epidemics and the need for expensive vaccines against diseases like influenza. Conversely, one may contend that it is the cells, rather than the animals, that proliferate in large quantities within incubators to generate cultured meat. Regrettably, our understanding of the full ramifications of meat culture on public health remains incomplete, given that in vitro meat is a novel product. Several authors argue that cell culture lacks proper regulation, which may lead to unexpected biological pathways (Chriki and Hocquette, 2020). Given the substantial proliferation of cells, it is probable that some disruption in cell regulation will occur, similar to what happens in cancer cells. However, it is possible to remove these deregulated cell lines for manufacturing or consumption purposes. Consuming in vitro meat may have unforeseen impacts on muscle structure as well as potential implications for human metabolism and health. Cattle face a significant challenge with antibiotic resistance. In a regulated setting, cultured meat enables vigilant oversight to promptly detect and prevent contamination. However, introducing antibiotics to avoid contamination, even if done sporadically to halt early contamination and disease, diminishes the persuasiveness of this argument (Welin, 2013).

Manipulating the types of fats utilized in the production medium can regulate the nutritional composition of cultured beef. Controlling the ratio between saturated fatty acids and polyunsaturated fatty acids is a straightforward task. Alternative fats, such as omega-3, can serve as substitutes for saturated fats, although it is crucial to manage the potential increase in rancidity. Researchers have devised novel approaches to enhance the concentration of omega-3 fatty acids in meat within the framework of existing animal production techniques (Ponnampalam et al., 2018). Furthermore, there is currently no established approach to enriching cultured meat with certain micronutrients found in animal products, such as vitamin B₁₂ and iron, which are known to promote positive health (Scollan et al., 2014). Furthermore, supplying any micronutrient in a suitable matrix can amplify its beneficial impact. Regarding in vitro meat, it remains uncertain whether the other biological substances and their arrangement within cultivated cells could enhance micronutrient benefits on human health. It is crucial to have a thorough understanding of the absorption of micronutrients (such as iron) by cultured cells. The health-promoting effects of micronutrients may decrease due to the content of the culture medium. Furthermore, the use of chemicals in the medium results in cultured meat that is more akin to “chemical” food, hence diminishing its clean-label status (Bhat et al., 2017). Alternative protein requires novel food approval for human consumption. When being fed to animals, a similar approval is needed, which in Europe is the EU feed catalogue EU 68/2013.

Cellular culture and tissue

The development phase of a mature meat production system involves initially gathering suitable cells capable of muscle formation. This is followed by their substantial proliferation in a bioreactor. Currently, there is no large-scale bioreactor specifically designed and made for dense meat production, even though muscle fiber strips can be generated. The muscular anatomy requires a mechanism like the circulatory system to transport essential nutrients and oxygen in close proximity to the developing cells or fibers and eliminate metabolic waste. Despite the successful application and examination of cutting-edge biomass technologies, researchers have not yet achieved the development of muscles with internal blood arteries, which would help with the provision of oxygen and nutrients, even though it is possible to generate minuscule muscle components that receive an ample supply of nutrients

and oxygen. In order to resemble actual flesh, cultured muscle tissue also requires muscular stretching. According to [Ostrovidov et al. \(2014\)](#), fibroblast stem cells arrange collagen or collagen matrix into fiber fibers, which are then anchored to produce tension in developing muscle fibers. It greatly enhances protein synthesis in cultured muscle. [Tomasch et al. \(2023\)](#) found that the cyclic tensile procedure employed in their research enhanced protein production. However, [Vandenburgh et al. \(1989\)](#) documented the beneficial impacts of cycle stretching on muscular development. In addition to proteins, various types of proteins present in the muscle play an important role in determining the color, texture, and flavor of meat. For example, the protein myoglobin is responsible for the red color of beef and serves as a significant iron source in meat. The process of muscle contraction promotes the production of myoglobin, which can enhance the pigmentation of meat. The most challenging aspect of growing meat lies in determining the optimal composition of the culture medium, despite the existence of numerous cell culture techniques ([Jandyal et al., 2021](#)).

Cultured meat classification

Various alternatives to traditional meat include plant-based and fungal options, as well as cultured or synthetic meat created by tissue engineering or derived from genetically modified organisms and cloned animals using genetic engineering techniques.

Concerning cultured meat, scientists cultivate cells in a laboratory environment by introducing a scaffold containing a limited quantity of muscle cells and subsequently promoting their growth through cell culture techniques. Culture medium composition, pH, temperature, and oxygen levels are critical for embryo development and viability ([Wale and Gardner, 2016](#)). Key factors affecting cell growth include moisture, light, nutrients, hormones, and sterile environments ([Segeritz and Vallier, 2017](#)). Cell culture stimulates cell development in an artificial environment using chemical and physical factors. Plant regeneration can also be achieved using this approach. Both cell sources have the ability to produce goods.

- (i) Primary cells that come from the original tissue or cell lines can divide in two ways: (1) induction, which involves changing the cells' genes or using chemicals to make them multiply endlessly, and (2) spontaneous mutations, which give the cells the ability to divide forever.
- (ii) Primary cells derived from intact tissue. Although muscle stem cells have received significant attention, there are other types of stem cells, such as mesenchymal stem cells, that can thrive in conditions without serum. Mesenchymal stem cells (MSCs) are a type of cell that are present in several organs and are responsible for forming connective tissue. MSCs are multipotent progenitor cells found in the connective tissues of various organs throughout the body ([Young et al., 1995](#); [Chen et al., 2024](#)). These plastic-adherent, fibroblastoid cells can differentiate into multiple mesodermal lineages, including bone, cartilage, fat, and muscle ([Alhadlaq and Mao, 2004](#); [Phinney and Prockop, 2007](#)). Initially isolated from bone marrow, MSCs have since been identified in numerous other tissues ([Chen et al., 2024](#)). This is because of their greater ability to reproduce rapidly. Embryonic stem cells, obtained from embryos in the early stages of development, exhibit continuous growth but present difficulties in their transformation into muscle cells. Furthermore, while it is possible to obtain human primary cells for the purpose of meat cultivation, growing human tissue for this goal would involve significant ethical, medical, and regulatory consequences ([Van Loo et al., 2020](#)).

Disadvantages of cultivation meat

There are, undoubtedly, those who oppose the development of laboratory-grown meat, each with their distinct reasons. Critics argue that this meat is generated by transgenic methods and raises concerns regarding transgenic products. Certain individuals are unable to tolerate the consumption of meat. Cultured meat lacks the authentic taste of natural meat due to its synthetic composition, which may not be appealing to meat enthusiasts seeking a genuine experience. Furthermore, a significant issue arises from the decline in employment opportunities associated with the rearing of cattle. The advent of lab-grown meat in the market leads to the displacement of conventional workers and ranchers, resulting in a loss of employment and income for them.

Concerns about the future of synthetic meats

High price of synthetic meat

Synthetic meat, or cultured meat, is being explored as a potential solution to meet growing protein demands and address environmental and ethical concerns associated with conventional meat production ([Bonny et al., 2017](#); [Muslu, 2022](#)). While it offers potential benefits, significant challenges remain. These include high production costs, achieving desired taste and texture, and scaling up to industrial levels ([Muslu, 2022](#); [Warner, 2019](#)). Consumer acceptance is mixed, with concerns about naturalness and price being major barriers ([Warner, 2019](#); [Wilks and Phillips, 2017](#)). Nevertheless, given the current high cost of these products, it appears unlikely that cultured meat will be able to displace animal products on the market and mitigate their associated drawbacks over the coming decades. In 2013, the world's first lab-grown hamburger was unveiled, marking a significant milestone in culinary history ([Noble, 2023](#)). It was created by Dutch researcher Mark Post at Maastricht University ([Bomgardner, 2018](#)). The burger, made

solely from muscle cells without fat, was described as “close to meat” but “not as juicy” by food experts (Noble, 2023). The event, held in London, was a carefully orchestrated media presentation aimed at attracting research funding (Kupferschmidt, 2013). Despite mixed consumer reactions, the potential animal welfare and environmental benefits of producing meat without raising and slaughtering animals were highlighted (Bomgardner, 2018). This breakthrough sparked the development of a new industry, with entrepreneurs working to make lab-grown meat a commercial reality (Bomgardner, 2018). A decade later, the prospect of lab-based food on dining tables seems slightly closer to fruition (Noble, 2023). Moreover, if animal-based food is eliminated from the food chain and lab-grown meat becomes obligatory, it will transition meat into a high-end commodity, potentially leading to nutritional crises among the human population (Singh et al., 2022). Despite these challenges, some consumers are willing to try synthetic meat, although fewer are ready to adopt it as a regular replacement for conventional meat (Wilks and Phillips, 2017). The future of synthetic meat remains uncertain, with plant-based alternatives potentially outpacing its development (Warner, 2019).

Economic, social and security consequences

During the second phase, if investors in this sector can effectively lower costs and receive government backing, the repercussions of this matter will manifest in the form of the decimation of the animal husbandry industry. These implications will encompass the disruption of the food by-product cycle, desertification, and the emergence of societal problems. The decline of the animal husbandry sector has resulted in significant detrimental effects on rural employment, potentially leading to the depopulation of suburban areas and non-industrial regions within the country. The phenomena of marginalization and widening wealth disparity in society will be a significant consequence of this issue. This is because production resources are controlled by a select few individuals who possess capital and resources. Encountering desolate regions and remote settlements will potentially raise security issues. This problem will be more severe in poorer nations (Sebo, 2018).

Loss of cultural diversity

Another issue about cultured meat pertains to its potential alignment with standardized production practices, such as those found in the fastfood industry. This could lead to a reduction in cultural variation in food. Food science specialists predict that diminishing the hedonic experience of consuming a diverse range of meals will be a significant cognitive and emotional obstacle for individuals in this scenario (Singh et al., 2022).

The serious damage of removing animal husbandry industry on other parts of human food supply chain

Conversely, the animal breeding sector utilizes the by-products of the human food industry, such as straw, bran, meal, or abattoir waste. The removal of consumers from these products leads to a rise in agricultural products and human food prices. However, reintegrating these waste materials back into the food cycle or natural environment necessitates the implementation of cost-effective measures. However, animals can transform these substances into high-quality nutritional molecules. Furthermore, animal dung holds a prominent position as an economical and valuable resource in the agriculture sector (Noble, 2023).

Demand-side issues

First and foremost, it is crucial to acknowledge that expanding the range of choices available to customers is widely considered advantageous for their overall well-being. Certain consumers may disregard the new product, but others may happily consume it. This is the conventional benchmark for decision theory. It assumes that increased flexibility does not result in confusion or regret due to inadequate information or behavioral factors. Therefore, we anticipate that lab-grown beef will enhance customers' well-being on a fundamental level. It is critical to note that it may also provide vegetarians with the possibility of eating a substance resembling “meat” (Sebo, 2018).

It is important to note that traditional meat and cultured meat have significantly different production methods. However, they may or may not also vary in taste and characteristics. As previously indicated, technology may not be capable of precisely replicating the specific conventional meat choices that consumers desire. Technological advancements have enabled the generation of various types of meat, including “profiled meat,” which can be modified to possess healthier lipid content, B₁₂ vitamin, and other desirable qualities. This may make it more appealing to certain consumers than traditional meat. The agro-food sector commonly presents this argument concerning processed food fortified with vitamins, micronutrients, and similar substances. Furthermore, consumers are expected to experience enhanced well-being due to this increased option flexibility (Peker et al., 2024).

The absence of a market for cultured meat now precludes an accurate assessment of its demand, limiting the ability to make precise determinations and instead relying on forecasts. Previously, demand-side research focused on investigating consumers' approval through experimental, survey, or focus group investigations. Stephens et al. (2018) highlight the wide range of reactions observed both within individual research and across different investigations. Our primary source of information is the meta-analyses conducted by Bryant and Barnett in 2018 and 2020 (Bryant and Barnett, 2018). According to Hocquette et al. (2015) and Wilks and Phillips (2017), the percentage of participants who are open to trying cultured meat differs widely, with proportions ranging from 5% to 11% in the first study and up to 65% in the latter. Individuals who exhibit an increased inclination

to embrace innovation tend to possess higher levels of education, be younger, and hold more progressive political beliefs, on average (Post and Hocquette, 2017; Wilks and Phillips, 2017). Esthetic issues related to taste, texture, and appearance significantly influence the acceptance levels of cultured meat. Participants also expressed a range of additional worries, such as the perceived lack of naturalness, feelings of revulsion, concerns about safety and health, and the expected cost (Wilks and Phillips, 2017).

Several studies investigate the precise aspects that specifically impact consumers' acceptability. Weinrich et al. (2020) discovered in a recent study that ethics is the predominant motivator. Surprisingly, providing participants with information on the technical aspects of cultured meat production may not lead to an increase in acceptance. A significant determinant that diminishes acceptability is the "naturalistic heuristic," which refers to the conviction that "what is natural is commendable" (Sebo, 2018). In their study, McDonald and Vivalt (2018) examine the influence of debunking messages on this cognitive shortcut, concluding that the most effective approach is to advocate for accepting or embracing concepts that are considered abnormal rather than denying their existence (McDonald and Vivalt, 2018). Furthermore, they imply that efforts opposing cultured meat may effectively reduce acceptability, as these messages have a strong and lasting influence. Bryant and Dillard (2019) and Bryant et al. (2019) discovered that presenting the idea that conventional meat is unnatural led to a noteworthy rise in certain indicators of acceptance for lab-grown meat when compared to alternative messaging. Furthermore, it is noteworthy to observe that several participants in the tests exhibit an initial response of repulsion. Hopkins and Dacey (2008) argue that cultural factors predominantly influence disgust. One possible reason for this could be the perception of immorality associated with the innovation. Nevertheless, we will subsequently contend that this moral argument lacks a firm foundation and is therefore prone to diminishing in significance over time. As individuals learn more about cultured meat and its production process, neophobia (the unwillingness to try new things) can be overcome. Nevertheless, Bryant and Barnett (2018) indicate that there are also indications of wider community apprehensions around the decline of traditional animal agriculture, skepticism toward corporations involved in cultured meat production, and the substantial energy demands associated with its production. According to the studies conducted by Wilks and Phillips (2017), Slade (2018), and Tucker (2014), there is evidence suggesting that, on average, males exhibit a higher level of willingness to experiment with cultured meat (Wilks and Phillips, 2017). The gender disparity in sentiments toward conventional beef may initially come as a surprise, given the abundance of evidence suggesting that males generally hold more favorable views toward it. It is true that males generally consume a greater amount of meat and are less likely to engage in campaigns promoting meatless days. Moreover, consuming meat is associated with the perception of masculinity and the exertion of dominance. One plausible hypothesis is that males are usually more receptive to adopting new ideas, as observed with GMO products (Post and Hocquette, 2017). An alternative explanation could be that females exhibit a greater level of concern over potential health risks linked to novel food items. Females typically assume the responsibility of procuring food-related products for the entire household, which may contribute to this explanation. Additionally, they may exhibit heightened sensitivity toward scientific ambiguity and a greater inclination to embrace vegetarian cuisine as a substitute for traditional meat.

An inherent constraint of the current studies on customers' acceptance is the lack of investigation into price-related impacts, such as consumers' willingness to pay (WTP) or demand at different price levels. Van Loo et al. (2020) conducted a study of over 1800 US consumers who participated in a choice experiment, which was an exception to the usual practice (Van Loo et al., 2020). Participants were given the choice between conventionally farmed beef, a pea protein-based plant burger, an animal-like protein-based plant burger generated from yeast, cultured meat, or the option of not buying any of the goods. When participants were asked to select a food product with constant prices, 72% of survey respondents chose farm-raised beef, while only 5% opted for cultured meat (Van Loo et al., 2020). The majority of participants still went for traditional beef, despite the availability of substantial savings. However, individuals who follow a vegetarian diet, males, and younger and more highly educated individuals show a stronger preference for cultured meat over farm-raised cattle. Two further recent articles based on willingness-to-pay (WTP) analysis also indicate that consumers predominantly reject cultured beef, with comparable findings (Stephens et al., 2018).

Highlighting the methodological constraints of current studies on customer acceptance is important. Crucially, these studies are theoretical and hence inherently encounter conventional challenges related to hypothetical bias in survey surveys. Rolland et al. (2020) conducted a study where they eliminated a key hypothetical factor by notifying consumers that they were consuming conventional meat, even though they believed they were tasting true cultured meat. Put simply, they employed deceit in their sampling scenario. Remarkably, the researchers discovered that every participant sampled the "cultured" hamburger and universally judged its flavor to be superior to the traditional one, despite the lack of any discernible objective distinction. Additionally, they discovered that prior knowledge of lab-grown meat is the most accurate indicator of its acceptance, underscoring the significance of being familiar with the concept. This highlights another inherent drawback of consumer acceptability surveys, which is the unfamiliarity of the situation. The survey participants were completely unfamiliar with the food product, and some of them may have never even heard of it (Pakseresht et al., 2022).

The presence of hypothetical bias and a lack of familiarity may compromise the reliability of consumers' acceptance studies. The specific wording and presentation greatly influence the outcomes of research, leading to significant variations. Predicting future attitudes toward this breakthrough, as well as the future social norms and political context during the commercialization of cultured meat, presents inherent challenges. It is plausible that omnivores may encounter growing societal pressure in the next decade to decrease or abstain from consuming meat as more animal-friendly alternatives become accessible. The availability of cultured meat or other protein alternatives in restaurants, if consistently included on menus, might potentially raise awareness of the issue and gradually shift societal attitudes (Welin, 2013).

Curiously, Bryant et al. (2019) and Bryant and Dillard (2019) discovered that individuals' affinity for meat is a reliable indicator of their willingness to embrace lab-grown meat, but it does not have the same effect on their acceptance of plant-based meat. This

suggests that cultured meat may have an increased capacity to replace the demand for individuals who consume large amounts of meat and are not interested in plant-based alternatives. An unresolved inquiry pertains to the forthcoming stance of individuals who follow a vegetarian or vegan lifestyle. Vegetarians in affluent countries range from 3% to 10% of the population. The data indicates a noticeable upward trend, particularly among the younger demographic. How will vegetarians respond to widespread cultured meat options? The majority of vegetarians may exhibit indifference and continue to adhere to their existing dietary practices. However, this is contingent upon the underlying motive behind vegetarianism adoption. Some individuals may find the prospect of consuming meat without causing harm to animals enticing. However, others may reject cultured meat due to uncertainties about its health impacts or the energy-intensive and technical nature of its production (Stephens et al., 2018).

This discussion regarding the demand from specific consumer segments and the impact of food substitution offers a crucial inquiry into the economic aspects of cultured meat. One might hypothesize and envision multiple possibilities that could influence future meat markets. One scenario pertains to a circumstance in which cultured meat companies successfully manufacture economically viable products, particularly minced meat analogs. These goods are likely to appeal to avid users of (inferior) meat products, particularly those derived from factory farming. These items have the potential to gain substantial market dominance in emerging economies. Therefore, this situation could have significant implications for the meat production industry, as well as substantial health, environmental, and ethical ramifications (Sebo, 2018). Another situation to consider is a “niche” market. If cultured meat products continue to be significantly costlier than conventional goods, they are likely to appeal only to particular consumer groups, such as flexitarians or vegetarians, as well as those who are more conscious of the indirect effects of their consumption and less bothered about the intricacies of the production process. Similarly, on rare occasions or for specific recipes, certain food products should be reserved, like cultured foie gras. The ramifications of this scenario for society may be minimal.

Additional intricate or nuanced inquiries emerge. Sebo (2018) examines the topic of identity. He contends that numerous individuals perceive their gender identity, sexual orientation, cultural identity, religious identity, national identity, professional identity, and other aspects as closely linked to the act of consuming meat. If this idea proves to be accurate and supported by facts, individuals will need to contemplate whether their habits and traditions align with cultured meat. If not, what possible solutions could address this conflict? Sebo (2018) further underscores that in the face of disturbances, individuals tend to respond defensively, striving to maintain known patterns of cognition, emotion, and behavior.

Consumers may develop self-serving ideas to alleviate the guilt they experience from consuming meat, as suggested by the meat paradox hypothesis. Essentially, individuals tend to consume animals while avoiding acknowledgment of their role in the pain and killing of animals. They may unconsciously seek reasons and rationalizations to alleviate their sense of responsibility. The usual models of cognitive dissonance predict that the mere presence of meat substitutes, such as cultured meat, could potentially enhance ethical attitudes toward the environment and animals. Vivalt and Macdonald (2018) generally reject this idea, unless they adequately account for the “naturalistic heuristic.” Specifically, the findings demonstrate that those who hold a favorable view of cultured beef do experience a shift in their ethical convictions, aligning with the principles of cognitive dissonance theory.

This hypothesis highlights the intricate and varied psycho-sociological views toward meat. According to Vialles’ (1988) anthropological research, there is a distinction between “zoophages” (individuals who openly acknowledge their consumption of animals) and “sarcophages” (individuals who prefer to consume non-animal-related products) (Vivalt and Macdonald, 2018). Based on prior knowledge, cultured meat may satisfy the second group of people but not the former, unless there is an effective symbolic substitution, as mentioned in Burgat et al.’s (2022) discussion. Burgat suggests that cultured meat prompts inquiry into the true nature of meat and that people may continue to perceive the concept of a meat-centered diet even after conventional meat has been replaced by cultured alternatives (Burgat et al., 2022). Van der Wee and Driessen (2019) argue that as the definition of meat becomes ambiguous, the significance and sensations associated with consuming it become uncertain. The major focus of this technique is the ambiguity and ambivalence surrounding both traditional and cultured meat. This technique offers supplementary justifications for the constraints of research that forecasts the preferences of future consumers.

Conclusion

The rising demand for protein for food cannot be met sustainably by meat. A viable solution to the substantial adverse impacts of existing meat production on the environment and human health can be found in the cultivation of lab-grown meat, a revolutionary procedure to provide protein at scale and with reduced environmental footprint. Cultivating meat is a crucial endeavor that has the potential to meet the dietary needs of the future. The occurrence of previous and ongoing diseases, coupled with the growing worldwide demand for meat, has highlighted the pressing necessity to explore alternate solutions to animal production. Environmental campaigners caution that the increasing global consumption of meat, particularly in major fast-growing nations like China, is unsustainable due to the higher resource requirements of beef, but also other types of meat like poultry compared to plants. Cows emit significant quantities of greenhouse gases on their own, contributing to the exacerbation of global warming. Laboratory cultivation has the potential to be a groundbreaking solution for both food production and environmental sustainability if it becomes economically viable. This type of meat minimizes the suffering of animals used for human consumption and also has a substantial impact on reducing greenhouse gas emissions. The meat, also known as a meat substitute or meat imitation, closely resembles the texture, taste, look, and chemical qualities of real meat. The “alt protein” market is huge and growing fast. Some of these imitation meat products are derived from soybeans or gluten. Vegetarians are one of the pseudo-meat consumers. Experts anticipate that lab-grown meat, commonly known as cultured meat, will be available for purchase at comparable prices to

conventional animal meat, with price parity being achieved within a few years according to predictions. Also, single cell protein (SCP) holds a strong promise, for feed and food. All of them are subsumed as “cellular agriculture.” However, experts caution that significant advancements are still required before this technology can fully realize its potential, to reach price parity with meat and be ready for scale-up.

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